

Water Tank Optimization

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Mathematical Optimization

Universe

- Open Top: $\{\text{radius}_m, \text{height}_m, \text{surface_area}_m, h_{\text{over}_r}, \text{radius_symbolic}, \text{height_symbolic}\}$
- Closed Top: $\{\text{radius}_m, \text{height}_m, \text{surface_area}_m, h_{\text{over}_r}\}$
- Cube Comparison: $\{\text{side}_m, \text{area_open}_m, \text{material_savings_pct}\}$

Variables

Symbol	Meaning	Type / Range
x	decision variable	$\mathbb{R}$

Structure

Design an open-top cylindrical tank to hold 1000 liters using the least material (minimize surface area). The tank consists of a circular base and a cylindrical wall with no lid. Given a required volume V , find the radius r and height h that minimize the total sheet-metal area $S = \pi r^2 + 2\pi r h$ subject to the volume constraint $\pi r^2 h = V$.

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

- | | | |
|-----|---------------------------------|---|
| (1) | Optimization via calculus | minimizing a function subject to an equality constraint |
| (2) | Derivatives and critical points | setting $dS/dr = 0$ to find candidate optima |
| (3) | Second derivative test | confirming $d^2S/dr^2 > 0$ at the critical point |
| (4) | Lagrange multipliers | equivalent formulation for constrained optimization |
| (5) | Geometric insight | the optimal open-top cylinder has $h = r$; the closed-top one has $h = 2r$ |

Approach

Suggested approachSymbolic Differentiation + Critical Point Analysis (SymPy)

Complexity class:

Available tools: Symbolic Differentiation + Critical Point Analysis

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	Symbolic Differentiation + Critical Point Analysis (SymPy)
Time:	0.2833s

Solution Details

Method: Symbolic differentiation and critical point analysis via SymPy. The volume constraint $V = \pi r^2 h$ is used to eliminate h from the surface area expression $S(r, h)$, yielding a single-variable function $S(r)$. Taking dS/dr and setting it to zero gives the optimal radius. The second derivative test confirms the critical point is a minimum. The closed-top variant ($S = 2\pi r^2 + 2\pi r h$) is solved analogously for comparison.

Verification

- ✓ Volume Constraint Satisfied Passed
- ✓ Surface Area Recomputed Matched
- ✓ Second Derivative Positive Passed
- ✓ First Order Condition $dS/dr = 0$ Passed
- ✓ Open Top H Equals R Passed
- ✓ Closed Volume Constraint Passed
- ✓ Closed Surface Area Recomputed Passed
- ✓ Closed Top H Equals $2R$ Passed
- ✓ Perturbation $R+0.001$ Gives Worse S
- ✓ Perturbation $R-0.001$ Gives Worse S

3. Interpretation

The Question

Design an open-top cylindrical tank to hold 1000 liters using the least material (minimize surface area).

The Answer

A feasible solution was found.

What This Means

- Symbolic optimal radius: $r^* = (V / \pi)^{1/3}$ for open-top
- Symbolic optimal height: $h^* = (V / \pi)^{1/3} = r^*$ (i.e., height equals radius for open-top)
- Numerical surface area for $V = 1000 \text{ L (1.0 m}^3\text{)}$: approximately 5.536 m^2
- Closed-top comparison: $r^* = (V / (2\pi))^{1/3}$, S slightly larger
- Material savings versus a cube of equal volume

Symbolic differentiation and critical point analysis via SymPy. The volume constraint $V = \pi r^2 h$ is used to eliminate h from the surface area expression $S(r, h)$, yielding a single-variable function $S(r)$. Taking dS/dr and setting it to zero gives the optimal radius. The second derivative test confirms the critical point is a minimum. The closed-top variant ($S = 2\pi r^2 + 2\pi r h$) is solved analogously for comparison.

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution